Very little experimental work has been carried out on this molecule; I have found only that of Hirt, Halverson and Schmitt[3], which reports the lowest (singlet) transition as having an energy of 5·62 eV. and a symmetry $^1A_2'$. Further experimental work is required to confirm our assignment.

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Remanent Magnetization of Lower to Middle Eocene Basalt Flows from Oregon

The Siletz River Volcanic Series of north-western Oregon offers an ideal opportunity to determine the direction of the Earth's magnetic field in the western United States during the early Tertiary. The formation consists of 3,000–5,000 ft. of mainly submarine basaltic flows, flow breccia and tuff. Interbedded clastic sedimentary strata have yielded more than thirty species of fossils including brachiopoda and *Turritella andersoni* cf. subsp. *susanae* Merriam, on the basis of which J. Durham correlated this formation with the Capay shale of California[1]. Additional faunal assemblages, including some foraminifera collected subsequently to those referred to above, confirm this correlation and clearly establish the age of the Siletz River Volcanic Series as early middle Eocene to lower Eocene. Petrographically, the flows are porphyritic in texture, with phenocrysts of plagioclase and augite in a ground mass which consists of volcanic glass, plagioclase laths, and granules of augite and magnetite, the latter constituting as much as 15 per cent of the rock[1].

Fifty-seven oriented samples were collected from eight flows distributed across the coast range at a maximum distance of thirty-eight miles apart. All sampling was done in quarries or deep road cuts to avoid the effect of the weathered zone. Accurate geological attitudes could be measured in the associated sedimentary beds.

Considerable scatter was encountered within individual flows. This was strikingly reduced by heating the specimens to 225° C. and cooling them in a magnetic field of zero intensity—a procedure used by Doell at lower temperatures to reduce scatter in sediments[2]. Figs. 1 and 2 show the same twelve specimens which were cut from twelve oriented samples from a single flow. The convention used is that all points are plotted on the lower hemisphere of an equal area projection, solid points representing positive inclinations, that is, north-seeking poles downward, and open circles negative inclinations. As an individual specimen was heated to progressively higher temperatures, the direction of magnetization was found to move roughly in a great circle connecting the direction of the present dipole field and the mean direction of the flow. The heating process may either chemically destroy alteration minerals which possess a crystallization magnetization[2], or it may remove the component due to viscous magnetization.

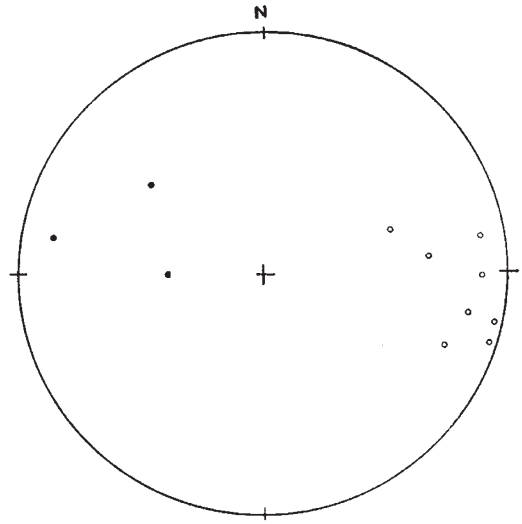


Fig. 1. Original direction of magnetization of twelve samples from flow *A*

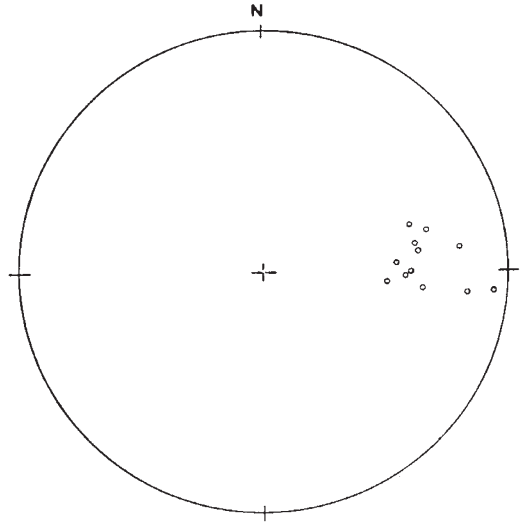


Fig. 2. The same twelve samples of Fig. 1 after heating to 225° C. and cooling in zero field

In one flow four samples were collected across a single pillow in which well-developed columnar joints radiated from the centre. Virtual elimination of a considerable initial scatter on heating suggests that any distortions in the magnetization which might possibly have been introduced by the stresses active in producing columnar jointing were removed by the heating process.

The mean direction for each of the eight flows before correcting for tilt is plotted in Fig. 3. Three of the flows show positive and five negative inclinations. There is no obvious petrographic difference between the antiparallel sets. By Graham's classic test for magnetic stability, the scatter should be reduced on correcting for tilt[3]. Fig. 4 shows this to be the case, all flows moving toward a common centre except flow *H*. This flow also has the greatest circle of confidence of any flow.

Since a single flow cools rapidly with respect to the rate of secular variation and since there are

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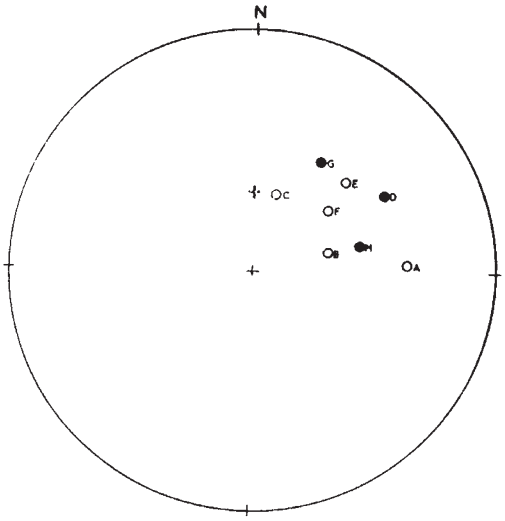


Fig. 3. Mean directions of eight flows before correcting for tilt. Cross indicates position of present dipole field

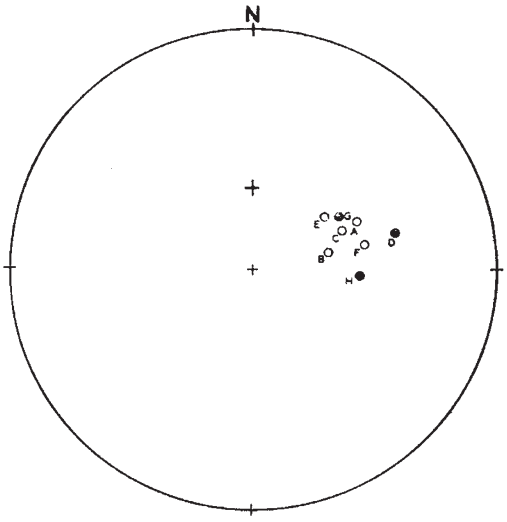


Fig. 4. Mean direction of eight flows after correcting for tilt

hundreds of feet of sediments associated with the flows, the eight flows may be considered a sampling which is random with respect to the secular variation, and Fisher's statistical method applied to give a mean direction of north 70° east and 55° down, with $\lambda_{95} = 7°$. (Fisher's model is not rigorously applicable to the analysis of a set containing anti-parallel directions, but is used here for comparative purposes.) The combined fifty-seven samples may also be treated by Fisher's method to give a direction of north 67° east and 56° down, with $\lambda_{95} = 3\frac{1}{4}°$. The former is regarded as the more valid procedure, and the field thus computed is such as would be produced by an axial dipole field with a pole at latitude 37° north and longitude 50° west. This pole lies about 30° north-east of the average pole found by Clegg *et al.* from the Deccan Traps[4], which are probably slightly older. It differs by a greater distance from the Eocene poles computed from European data[5].

These measurements suggest the desirability of additional measurements in the early Tertiary in order to establish the rate of variation of the average field. From this we may be able to determine the accuracy in dating which is necessary for inter-continental comparisons of directions of remanent magnetization.

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Variation with Pressure of the Permittivity of Polythene

In connexion with the recent letter from Reitzel[1], some measurements made in 1954 of the variation with pressure of the permittivity of polythene may be of interest. For these measurements, disks of polythene carried aluminium electrodes (which had been evaporated on) and were immersed in oil to which hydrostatic pressure, of up to 600 kgm./cm.^{2}, was applied. One electrode on each disk was of smaller diameter than the disk itself, so that most of the stray capacitance at the edge of the disk was through polythene subjected to the same pressure as the rest of the specimen.

The results showed that the effect in various types of polythene, and mixtures with small amounts of other polymers, was similar; and that the main change of permittivity is that caused by the compression of the material. The remaining effect, interpreted as a variation in the polarizability α, is given by

$$\frac{1}{\alpha}\frac{d\alpha}{dP} = \frac{3\varepsilon}{(\varepsilon-1)(\varepsilon+2)}\frac{1}{C}\frac{dC}{dP} - \varkappa\left[1 - \frac{\varepsilon}{(\varepsilon-1)(\varepsilon+2)}\right]$$

where C is the capacitance of a sheet specimen; or, for polythene, in which $\varepsilon = 2{\cdot}29$,

$$\frac{1}{\alpha}\frac{d\alpha}{dP} = 1{\cdot}24\,\frac{1}{C}\frac{dC}{dP} - 0{\cdot}59\,\varkappa$$

The term $\frac{1}{C}\frac{dC}{dP}$ was found to be $12{\cdot}2 \times 10^{-6}$ per kgm./cm.^{2}; the compressibility of polythene at these pressures was taken as 31×10^{-6} per kgm./cm.^{2} The term $\frac{1}{\alpha}\frac{d\alpha}{dP}$ is therefore -3×10^{-6} per kgm./cm.^{2}, somewhat less than the earlier values mentioned by Reitzel; but as it arises only as the difference of two much larger terms, and as the compressibility of polythene is known to vary with pressure, the accuracy of this result cannot be good.

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